Spectral Bounds for Classical & Quantum Graph Parameters

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Spectral graph theory

- Spectral graph theory is a well-established branch of graph theory.
- It studies properties of a graph with the help of eigenvalues and eigenvectors of matrices associated to the graph such as the adjacency and Laplacian matrices.
- One of the most well-known results in graph theory is the Hoffman lower bound on the chromatic number.

Graph parameters I
Chromatic numbers and Lovász theta functions

- $\omega(G)$ – clique number
- $\chi_{v}(G)$ – vector chromatic number
- $\chi_{sv}(G)$ – strict vector chromatic number
- $\chi_{rv}(G)$ – rigid vector chromatic number
- $\chi_{\mathrm{vect}}(G)$ – vectorial chromatic number

- $\theta'(\overline{G})$ – Schrijver variant of Lovász theta function
- $\theta(\overline{G})$ – Lovász theta function
- $\theta^{+}(\overline{G})$ – Szegedy variant of Lovász theta function

- These parameters, other than $\omega(G)$ can all be calculated using semi-definite programming

Graph parameters II

- $\xi_f(G)$ – projective rank
- $\xi(G)$ – orthogonal rank
- $\xi'(G)$ – normalized orthogonal rank
- $\chi_q(G)$ – quantum chromatic number
- $\chi_q^{(1)}(G)$ – rank one quantum chromatic number
- $\chi_f(G)$ – fractional chromatic number
- $\chi_c(G)$ – circular chromatic number
- $\chi(G)$ – chromatic number

Hierarchies of graph parameters

$$\begin{tikzcd}
 & & & & \lceil \theta^{+}(\overline{G}) \rceil = \chi_{\mathrm{vect}}(G) \\
\omega(G) \arrow[r] & \chi_{v}(G) = \theta'(\overline{G}) \arrow[r] & \chi_{sv}(G) = \theta(\overline{G}) \arrow[r] & \chi_{rv}(G) = \theta^{+}(\overline{G}) \arrow[r] \arrow[ur] & \xi_{f}(G)
\end{tikzcd}$$

$$\begin{tikzcd}
\chi_{\mathrm{vect}}(G) \arrow[r] \arrow[dr] & \xi(G) \arrow[dr] & & & \\
\xi_{f}(G) \arrow[r] \arrow[ur] \arrow[dr] & \chi_{q}(G) \arrow[r] & \chi_{q}^{(1)}(G) \arrow[r] & \xi'(G) \arrow[r] & \lceil \chi_{c}(G) \rceil = \chi(G) \\
 & \chi_{f}(G) \arrow[r] & \chi_{c}(G) \arrow[urr] & &
\end{tikzcd}$$

Here $\longrightarrow$ is used to indicate the less or equal relationship.

Coloring & chromatic number

- We consider only undirected graphs without multiple edges and without loops.
- A c-coloring of a graph $G = (V, E)$ is a collection of colors

$$\{c_v : v \in V\}$$

  in $[c] = \{0, \ldots, c-1\}$ such that for all $vw \in E$

$$c_v \neq c_w.$$

- The chromatic number χ of a graph G is the minimum c admitting a c-coloring of G.
- The chromatic number is NP-hard to approximate.

Quantum coloring, quantum chromatic number

- A quantum c-coloring of a graph $G = (V, E)$ is a collection of orthogonal projectors

$$\{P_{v,k} : v \in V, k \in [c]\}$$

acting on $\mathbb{C}^d$ for some $d > 0$ such that
  - for all vertices $v \in V$

$$\sum_{k \in [c]} P_{v,k} = I_d \qquad \text{completeness}$$

  - for all edges $vw \in E$ and for all $k \in [c]$

$$P_{v,k} P_{w,k} = 0_d \qquad \text{orthogonality}$$

- The quantum chromatic number χ_q is the smallest c for which the graph G admits a quantum c-coloring.

Classical coloring as a special case of quantum coloring

- Let $\{c_v : v \in V\}$ be a collection of colors in $[c]$ that defines a classical c-coloring.
- For $v \in V$, consider the "*one-hot-encoding*"

  $$\{P_{v,k} : k \in [c]\}$$

  of the color $c_v \in [c]$, that is,

  $$P_{v,k} = [c_v = k],$$

  where $[\cdot]$ denotes the Iverson bracket.
- For instance, if $c = 3$ and some vertex v is colored with color $c_v = 2$, then the corresponding one-hot-encoding is given by

  $$P_{v,0} = 0, \quad P_{v,1} = 0, \quad P_{v,2} = 1.$$

Classical coloring as a special case of quantum coloring

- It is immediate that a classical c-coloring gives rise to a quantum c-coloring coloring in dimension $d = 1$.
- The orthogonal projectors in the resulting quantum c-coloring are the scalars 0 and 1.

Exponential separation between classical and quantum chromatic numbers

- The orthogonality graph, $\Omega(n)$, has the vertex set of ± 1-vectors of length n, with two vertices adjacent if they are orthogonal.
- The quantum chromatic number is linear in n (equal to n), whereas the classical chromatic number $\chi(\Omega(n))$ is exponential in n.

Hoffman lower bound on the quantum chromatic number

- Let $A = (a_{vw}) \in \mathbb{C}^{n\times n}$ be the adjacency matrix of a graph G, that is,
  - $a_{vw} = 1$ for all $vw \in E$
  - $a_{vw} = 0$ otherwise
- Let $\mu_{\max}$ and $\mu_{\min}$ denote the maximum and minimum eigenvalues of A, respectively.
- The Hoffman bound states

$$\chi_q \geq 1 + \frac{\mu_{\max}}{|\mu_{\min}|},$$

Quantum coloring $\Rightarrow$ matrix equality for the adjacency matrix

- We will show later that starting from a quantum c-coloring we can construct a unitary matrix

$$U \in \mathbb{C}^{n \times n} \otimes \mathbb{C}^{d \times d}$$

such that the following matrix equality holds

$$\sum_{\ell=1}^{c-1} U^{\ell}(-A \otimes I_d)(U^{\dagger})^{\ell} = A \otimes I_d.$$

- Once we have established the above matrix equality for the adjacency matrix, we can use results from majorization theory to prove the Hoffman bound.

Majorization inequality for maximum eigenvalues

- Let X and Y be two arbitrary hermitian matrices in $\mathbb{C}^{m\times m}$.
- It is well known that the maximum eigenvalues of the matrices satisfy

$$\mu_{\max}(X) + \mu_{\max}(Y) \geq \mu_{\max}(X + Y).$$

Proof of the Hoffman bound

- Applying (i) the majorization bound and (ii) the invariance of the spectrum under conjugation with unitaries to the matrix equality for the adjacency matrix

$$\sum_{\ell=1}^{c-1} U^{\ell}(-A \otimes I_d)(U^{\dagger})^{\ell} = A \otimes I_d.$$

  yields

$$(c-1)\mu_{\max}(-A \otimes I_d) \geq \mu_{\max}(A \otimes I_d).$$

- This further simplifies to

$$(c-1)|\mu_{\min(A)}| \geq \mu_{\max}(A),$$

  which is the Hoffman lower bound.

Inertial lower bound on the quantum chromatic number

- Let (n^+, n^0, n^-) denote the inertia of a graph G, that is, the number of positive, zero, and negative eigenvalues of its adjacency matrix A.
- The inertial bound states

$$\chi_q \geq 1 + \max\left\{\frac{n^+}{n^-}, \frac{n^-}{n^+}\right\}.$$

Rank inequality for positive semidefinite matrices

- A hermitian matrix $X \in \mathbb{C}^{m \times m}$ is called positive semidefinite if all its eigenvalues are nonnegative.
- Let $X, Y \in \mathbb{C}^{m \times m}$ be two hermitian matrices.

  We say that X is greater or equal to Y, denoted by $X \geq Y$, if their difference $X - Y$ is positive semidefinite.
- Assume that $X, Y \in \mathbb{C}^{m \times m}$ are two positive semidefinite matrices with $X \geq Y$.
- Then, we have the rank inequality

$$\mathrm{rank}(X) \geq \mathrm{rank}(Y).$$

Decomposition of the adjacency matrix in positive and negative parts

- Let $\mu_1 \geq \mu_2 \geq \ldots \geq \mu_n$ denote the eigenvalues of the adjacency matrix A and $P_1, \ldots, P_n$ the corresponding one-dimensional projectors.
- Define

$$A^+ = \textstyle\sum_{i:\lambda_i > 0} \mu_i P_i \qquad P^+ = \textstyle\sum_{i:\lambda_i > 0} P_i$$

$$A^- = \textstyle\sum_{i:\lambda_i < 0} |\mu_i| P_i \qquad P^- = \textstyle\sum_{i:\lambda_i < 0} P_i$$

- Then, we can write

$$A = A^+ - A^-$$

- Note that

$$n^{\pm} = \mathrm{rank}(A^{\pm}) \quad \text{and} \quad A^{\pm} = \pm P^{\pm} A P^{\pm}$$

and also $n = n^+ + n^0 + n^-$.

Proof of the inertial bound

- We start with the matrix equality

$$\sum_{\ell=1}^{c-1} U^\ell(-A \otimes I_d)(U^\dagger)^\ell = A \otimes I_d$$

- Plugging in $A = A^+ - A^-$ yields

$$\sum_{\ell=1}^{c-1} U^\ell \left((A^- - A^+\right) \otimes I_d)(U^\dagger)^\ell = (A^+ - A^-) \otimes I_d$$

- Multiplying both sides by P^+ from left and right yields

$$\sum_{\ell=1}^{c-1} P^+ U^\ell \left((A^- - A^+\right) \otimes I_d)(U^\dagger)^\ell P^+ = A^+ \otimes I_d$$

Proof of the inertial bound

- Dropping the subtraction on the left hand side yields

$$\sum_{\ell=1}^{c-1} P^+ U^\ell (A^- \otimes I_d)(U^\dagger)^\ell P^+ \geq A^+ \otimes I_d$$

- Using the rank inequality and some elementary results yields

$$(c-1) \cdot \mathrm{rank}(A^- \otimes I_d) \geq \mathrm{rank}(A^+ \otimes I_d)$$

  which simplifies to the inertial bound

$$c \geq 1 + \frac{n^+}{n^-}.$$

  The other inequality $c \geq 1 + n^-/n^+$ is proved similarly.

Proof of matrix inequality

- In the following, we will prove the matrix equality

$$\sum_{\ell=1}^{c-1} U^\ell(-A \otimes I_d)(U^\dagger)^\ell = A \otimes I_d$$

with the help of pinching and twirling.

Pinching

- Let $\{Q_k : k \in [c]\}$ be any collection of orthogonal projectors in $\mathbb{C}^{m\times m}$ that form a resolution of the identity.
- Then, the operation $\mathcal{C}$ that maps an arbitrary matrix $X \in \mathbb{C}^{m\times m}$ to

$$\mathcal{C}(X) = \sum_{k\in[c]} Q_k X Q_k$$

  is called pinching.
- We say that the pinching $\mathcal{C}$ annihilates X if

$$\mathcal{C}(X) = 0_m.$$

Pinching from quantum coloring

- Assume that the orthogonal projectors

$$P_{v,k} \qquad v \in V, k \in [c]$$

acting on $\mathbb{C}^d$ form a quantum c-coloring of G.

- Then, the block-diagonal orthogonal projectors that act on $\mathbb{C}^n \otimes \mathbb{C}^d$ and are defined by

$$\begin{aligned} P_k &= \bigoplus_{v \in V} P_{v,k} \\ &= \sum_{v \in V} |v\rangle\langle v| \otimes P_{v,k} \qquad k \in [c] \end{aligned}$$

give rise to a pinching $\mathcal{C}$.

Pinching from quantum coloring

- Due to the block diagonal structure of P_k we have

$$P_k(A \otimes I_d)P_k = \sum_{v,w \in V} a_{vw} |v\rangle\langle w| \otimes P_{v,k}P_{w,k}.$$

- Due to the orthogonality condition we have

$$a_{vw} = 1 \quad \Rightarrow \quad P_{v,k}P_{w,k} = 0_d.$$

- Therefore, this pinching annihilates $A \otimes I_d$, that is,

$$\mathcal{C}(A \otimes I_d) = 0_n \otimes 0_d.$$

Twirling

- Let $\{U_\ell : \ell \in [c]\}$ be any collection of unitary matrices in $\mathbb{C}^{m\times m}$.
- Then, the operation $\mathcal{D}$ that maps an arbitrary matrix $X \in \mathbb{C}^{m\times m}$ to
$$\mathcal{D}(X) = \frac{1}{c}\sum_{\ell\in[c]} U_\ell X U_\ell^\dagger$$
is called twirling.
- We say that the twirling $\mathcal{D}$ annihilates X if
$$\mathcal{D}(X) = 0_m.$$

Twirling from pinching

- Let $\{Q_k : k \in [c]\}$ be a collection of orthogonal projectors defining a pinching operation $\mathcal{C}$.
- Let $\omega = e^{2\pi i/c}$ be a cth root of unity.
- Then, the collection of unitary matrices $\{U^\ell : \ell \in [c]\}$ consisting of the powers of the unitary U

$$U = \sum_{k \in [c]} \omega^k Q_k,$$

defines a twirling operation $\mathcal{D}$ such that

$$\mathcal{C}(X) = \mathcal{D}(X)$$

for all matrices $X \in \mathbb{C}^{m \times m}$.

Twirling from quantum coloring

- Use the orthogonal projectors $P_{k,v}$, $v \in V, k \in [c]$, of a quantum c-coloring to construct the corresponding block-diagonal orthogonal projectors

$$P_k = \sum_{v \in V} |v\rangle\langle v| \otimes P_{v,k} \qquad k \in [c]$$

acting on $\mathbb{C}^n \otimes \mathbb{C}^d$ and the unitary

$$U = \sum_{k \in [c]} \omega^k P_k \, .$$

- Then, the twirling map $\mathcal{D}$ defined by the powers of U

$$\mathcal{D}(X) = \frac{1}{c} \sum_{\ell \in [c]} U^\ell X (U^\dagger)^\ell$$

annihilates $A \otimes I_d$. This is equivalent to the matrix equality.

Importance of twirling from quantum coloring

- The construction of twirling & pinching from quantum coloring leads to a unified way of proving the Hoffman and inertial lower bounds for χ_q.
- Moreover, any existing or future lower bound on the minimum number of operations for pinching and twirling to annihilate A is automatically a lower bound for χ_q.

Implication of the lower bounds

- The spectrum of the Clebsch graph on 16 vertices is $(5^1, 1^{10}, -3^5)$.
- It is known that $\chi = 4$.
- The Hoffman bound implies $\chi_q \geq 3$, whereas the inertial bounds $\chi_q \geq 4$.
- Sometimes these and other spectral lower bounds allow us to determine the quantum chromatic number.
- It is not known whether the quantum chromatic number is computable which is why spectral bounds are particularly useful and interesting.

Joint work with Clive Elphick

- P. Wocjan and C. Elphick, *Spectral lower bounds for the orthogonal and projective ranks of a graph*, arXiv:1806.02734, accepted for publication in Electronic Journal of Combinatorics.
- P. Wocjan and C. Elphick, *Spectral lower bounds for the quantum chromatic number of a graph*, arXiv:1805.08334.
- P. Wocjan and C. Elphick, *More Tales of Hoffman: bounds for the vector chromatic number of a graph*, arXiv:1812.02613.
- P. Wocjan and C. Elphick, *An inertial upper bound for the quantum independence number of a graph*, arXiv:1808.10820.